

Lent Week 5 & 6

lw703

1 Completed Task

In the experiment before, the model score is given as the optimal score:

```
def optimal_score(x, data, sigma=0.1):
    diffs = x.unsqueeze(1) - data.unsqueeze(0)
    log_gaussian = -0.5 * (diffs.clone().detach() ** 2)
    .sum(dim=-1, keepdim=True) / sigma ** 2
    weights = torch.softmax(log_gaussian, dim=1) # dim=-2, (N, K, 1)
    denoiser = (weights * data.unsqueeze(0)).sum(dim=1)
    score = (denoiser - x) / sigma ** 2
    return score
```

Here we train a model to fit it and it performs nearly the same as before.

Meanwhile, before the step size is set as a scalar, which undermines the representation ability of the model. Here we make the dimension of stepsizenet output same as the dimension of the data and now it can fit more complex data:

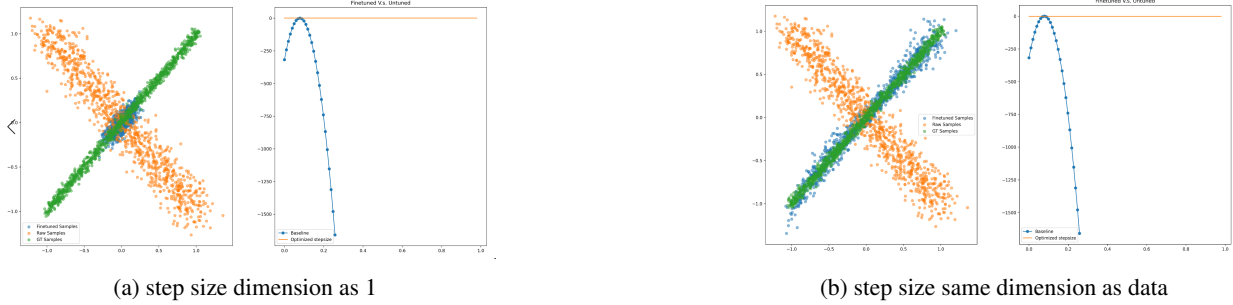


Figure 1: Generation results for 1-d and 2-d step size